

Millikan Oil Drops: Experiment II

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I. INTRODUCTION

Prior to discovery of the electron, electricity was thought to be a sort of fluid. For many years, electricity was an elusive phenomenon that was not well understood. Today, we explicitly manipulate electrons at a quantum scale to create previously unimaginable technologies like quantum computers and electron microscopes. The original oil drop experiments conducted by R.A. Millikan (see Ref. [1]) between 1909 and 1913 were seminal to this development over the last hundred years or so.

By the 1800's, scientists were already beginning to conceive of the electron's existence. Many thought there must be an "atom of electricity", which quantized the interactions facilitated by the electrical force. The English physicist JJ Thompson is credited with the actual discovery of the electron (1897) and an estimate of the charge to mass ratio, $\frac{e}{m}$. However, Millikan, inspired by previous attempts to measure the electron charge, was able to conduct his famous oil drop experiment in which he demonstrated this quantization and obtained an estimate of the electron's charge, e (see Ref. [2]). Millikan's findings put an estimate on the charge to be

$$e = 4.891 \times 10^{-10} \quad (1)$$

which was roughly equal to $e = 1.5924 \times 10^{-19}$ Coulombs, missing the currently accepted value by 0.6% (see Ref. [1]).

In replicating this experiment, it will be our goal to find similar results to the original Millikan findings and the accepted current day value of

$$e = 1.602176634 \times 10^{-19} \text{ C} \quad (2)$$

by measuring the terminal velocity of oil drops in the presence viscous drag, gravity, and electrostatic forces.

II. THEORY

The theory behind this experiment is fairly straightforward, and is adapted here from Ref. [2]. Atomized oil droplets are sprayed between the plates of a high voltage capacitor, gaining an electric charge, q , due to friction with the air. With the capacitor discharged ($V = 0$), the droplets only experience gravity, viscous drag, and negligible buoyancy forces. A high voltage on the capacitor creates an electric field within the region of interest which allows for the experimenter to manipulate the free fall velocity of the oil droplets. Thus, there are four forces of concern:

$$F_g = mg = \rho V_d g \quad (\text{gravity}) \quad (3)$$

$$F_v = 6\pi\eta av \quad (\text{viscous drag}) \quad (4)$$

$$F_B = \rho_{air} V_d g \quad (\text{buoyancy}) \quad (5)$$

$$F_E = qE \quad (\text{electrical force}) \quad (6)$$

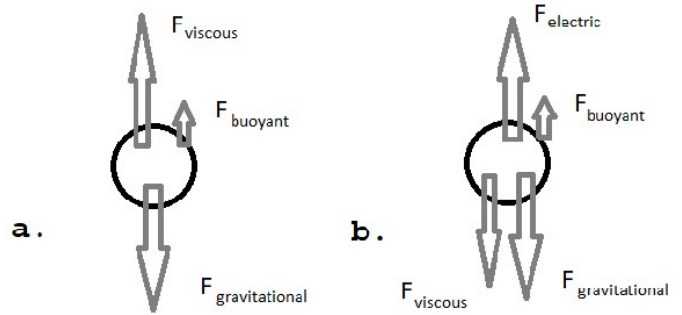


Fig. 1. Free-body diagram visualization of forces on oil droplets with (b) and without (a) the presence of an electric field. The buoyancy force is small enough to be ignored in both cases. The viscous drag force opposes the direction of motion. Figure adapted from Ref. [2].

where ρ is the density of the oil, V_d is the volume of the drop (which is not identical for all drops), η is the viscosity of air (a function of temperature), a is the drop radius, v is the drop velocity, q is the total electrical charge on the drop, and $E = V/d$ is the magnitude of the electric field between the plate capacitor at voltage V and plate separation d .

The buoyancy force is negligible because the density of the oil is much greater than the density of air ($\rho \gg \rho_{air}$, $F_g \gg F_B$).

Without an electric field, setting equations (3) and (4) equal and assuming terminal velocity, v_g , gives

$$mg = 6\pi\eta av_g \quad (7)$$

which allows a determination of the drop radius,

$$a = \left(\frac{9\eta v_g}{2g\rho} \right)^{1/2} \quad (8)$$

since drop radius is too small to be measured directly.

Without changing our assumptions, and accounting for the presence of an electric field, we balance the forces such that

$$qE = mg + 6\pi\eta av_E \quad (9)$$

where now we have v_E , which is the new terminal velocity of the drop. Equations (7) and (9) combine for a value of the charge on the drop based entirely on measurable quantities,

$$q = \frac{6\pi d}{V} \sqrt{\frac{9\eta^3}{2\rho g}} (v_E + v_g) \sqrt{v_g} \quad (10)$$

Importantly, we must correct the drag force (Stokes' Law) in the limit of extremely small spheres, which adds a factor

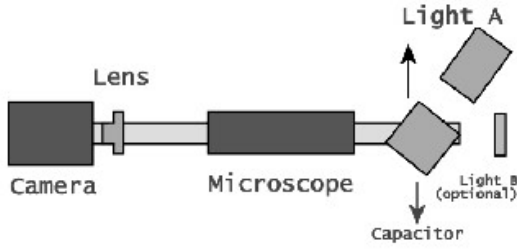


Fig. 2. Basic schematic of observation equipment setup, adapted from Ref. [2]. Atomizer and oil droplet delivery not depicted (see Fig. 3).

of $(1 + \frac{b}{aP})^{-1}$ where P is the atmospheric pressure and b is an empirical constant.

$$q = \frac{6\pi d}{V} \sqrt{\frac{9\eta^3}{2\rho g}} (v_E + v_g) \sqrt{v_g} \left(1 + \frac{b}{aP}\right)^{-1} \quad (11)$$

where $b = 6.17 \times 10^{-6} \text{ m} \cdot (\text{cmHg})$.

III. METHODS

The original experiment by Millikan was inspired by previous attempts to measure the charge of an electron using water droplets, though Millikan made a significant improvement to the experiment by using oil instead of water. The reason for the change being that water droplets would evaporate over the duration of the experiment. Evaporation made impossible the observation of a single drop for more than a minute, thus oil made a much better droplet candidate for precise observations.

The setup of the experiment involves a few materials:

- 1) Observation Camera / Lens
- 2) Microscope
- 3) Lighting for regions of interest
- 4) Parallel Plate Capacitor
- 5) High voltage power supply

The camera, lens, and microscope are aligned parallel to each other on an optical rail such that the oil droplets can be observed easily within the region of interest (between the plates of the capacitor charged by the high voltage power supply). An overhead view of the alignment is shown in Figure 2, as well as a depiction of the original experiment setup in Figure 3.

The microscope must be focused such that the measurement scale etched on the inside of the microscope is visible when looking through.

The capacitor itself is on a translation stage which allows for rotation of the plates as well as translation perpendicular to the optical rail, which facilitates the alignment process. The ideal angle of the capacitor is between 30 and 35 degrees relative to the optical rail, and the plates should be level. A wire should be inserted into the observation chamber while the camera is turned on to make sure the camera is correctly focused on the region of interest, and the microscope is aligned properly.

In the center of the top plate of the capacitor is a vent which allows for droplets of oil to enter the observation chamber between the plates. The atomizer should be used to spray droplets of oil above the capacitor such that they are in free fall when they enter the chamber.

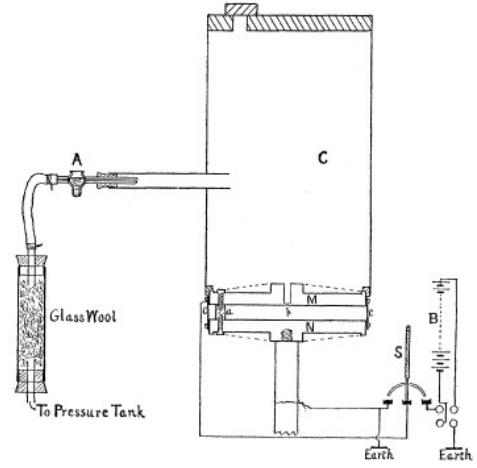


Fig. 3. Original Millikan Oil drop experimental setup. Atomizer (A) is shown to the left of the dust-free chamber (C) with the plate capacitor shown by M and N. Our experiment will use a similar design, but the atomizer is a handheld device with a small oil reservoir attached.

After calibrating the units of the etched scale, several videos of drops moving through the chamber should be recorded. In post, Tracker software can be used to plot the position of the oil droplets against time to calculate both v_E and v_g .

Choose slow moving drops to observe for the duration of the videos, as more time in the region of interest will allow for more data to be taken, generating more precise results for a particular drop.

IV. SAFETY CONSIDERATIONS

As the nature of the experiment requires high voltages to maximize the effects of the electric charge, proper safety procedures must be taken into consideration. At full power, the Matsusada ES Series power supply can generate a voltage of up to 1000 V at 3 mA. A shock from the power supply itself or the charged capacitor connected to it will certainly be both painful and dangerous.

Extreme care must be taken to avoid coming into contact with any of the high voltage equipment during operation. A few safety features are in place, including a large resistor which reduces the maximum current of the power supply, as well as a multimeter with an LED display connected to the capacitor so that the charge on the capacitor is always known. *These safety features alone are not enough to prevent you from receiving a painful shock.*

Note that even when the power supply is off, the capacitor can remain energized, and must be discharged completely (by connecting the top plate to ground) before it can be touched or before oil is sprayed into the chamber.

Unnecessary personnel should not be near the power supply or the capacitor when in operation.

V. ANALYSIS & INTERPRETATION

The experimental setup and data taking is fairly easy for this experiment, though data taking is time consuming. The real challenge in analysis will be the correct propagation of error. The final equation for the electric charge, q , includes

nine measurable quantities, most of which will have error that needs to be propagated. The value of q given will be the *total* charge on the drop. In order to show that charge is quantized, we must take enough measurements to determine that there are several discrete values which the charges seem to take consistently. This requires enough data for each value to show that it is not just a cluster. Additionally, the error propagation must yield final error bars on q which are small relative to the quantization gaps.

The best way to show the quantization will likely be a plot with q on one axis, and just about any other variable on the other axis. We expect the plot to look like many groupings of data points along vertical or horizontal lines with consistent spacing between them.

VI. CONCLUSIONS

Final conclusions must show that we can point to at least some consistent appearance of quantization of the charge q . This quantization can be used to determine the electron charge, e , which should be roughly the size of the gaps between charge "bands".

REFERENCES

- [1] R. A. Millikan, "The isolation of an ion, a precision measurement of its charge, and the correction of stokes's law," *Phys. Rev. (Series I)*, vol. 32, pp. 349–397, Apr 1911.
- [2] "Millikan Oil Drop Experiment v1.6," *CarmenCanvas*, 2024.